Vijay Ravi

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EDUCATION

University of Michigan **Expected: May 2027**
Bachelor of Science in Computer Science and Robotics (GPA: 3.81 / 4.0)

- Activities: MDP DIAG research lead, MFLY Autonomous sub-lead (former), Statistics & Data Science Grader (4 sem.)

EXPERIENCE

Siemens **May 2026 – August 2026**
Software Engineering Intern
Technologies: React, TypeScript, C++, C, MySQL, Agile, Microservices Architecture

- Piloted an LLM orchestrator that registers microservice agents as callable tools and routes user queries to them at runtime; adopted by four app teams during the pilot with no per-team changes.
- Rebuilt global navigation across server, client, and framework layers so group, role, and workspace switches commit in a single save, cutting a worst case of five render cycles down to one.
- Consolidated three separate HTML-injection Markdown renderers into one shared React component, shipped to four applications including Siemens Agentic Chat with copyable code blocks and LaTeX rendering.
- Prototyped a dashboard customization builder with persistence at user, group, role, and workspace scope.
- Wrote and shipped server-side POM queries that gather and filter Teamcenter business objects, with lookup caching.

Siemens **May 2025 – August 2025**
Software Engineering Intern
Technologies: React, C++, C, MySQL, Agile, Jest, Jenkins

- Shipped the personalized Teamcenter dashboard as a framework capability now used across every application in the product line: a configurable card schema backed by two new server endpoints and an extended personalization data model.
- Cut initial page-load API calls by up to 75% by virtualizing off-screen cards behind placeholder templates with configurable render-ahead, deferring card data fetches until scroll, owning through the entire software development lifecycle.
- Fixed a state-merge defect in the client framework's personalization data processor that was discarding incoming state.
- Partnered with 3 teams and 2 product owners through product demos to consolidate and deliver on technical requirements.

PROJECTS

Roblox Game Development **January 2024 – Present**
Technologies: TypeScript, Lua, REST, Networking, React, Git, Distributed client-server systems

- Architected client-server systems for 10+ games with 10M+ player visits and peaks near 1,000 concurrent players, covering networking, reusable services, React interfaces, and state machines
- Built fault-tolerant purchase persistence over a cached player data layer so transactions survive session and server failures.
- Secured server endpoints with authentication, request validation, rate limiting, and caching against malicious clients.
- Optimized high-volume messaging with type-safe protocols and buffer-based serialization; tuned memory and frame time.

Helivox **January 2021 – November 2023**
Technologies: Java, Spring Boot, Angular, Firebase, Apache Spark, Kubernetes, TypeScript, GCP, Git

- Architected Spring Boot microservices on Kubernetes for authentication, analytics, and content management.
- Built role-based UI generation and pagination producing 100+ configurable components for 200+ users.
- Implemented tiered authentication across client and server with role-based authorization for five staff roles, storing credentials and PII on Google Cloud and Firebase.

TECHNICAL SKILLS

Programming Languages: Java, SQL, C++, C, C#, Python, JavaScript, TypeScript, Lua, Linux Shell
Frameworks and Tools: React, Angular, Node.js, Spring Boot, Docker, Kubernetes, GCP, MySQL, OracleDB, Jest, Jenkins, Git
Concepts: Microservices Architecture, REST, CI/CD, Test Automation, Agile, Object-Oriented Design, DevOps, Data Structures
Certifications: Google Certified Professional Machine Learning Engineer